Mechanical waves and distant effects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use travel-time reasoning while distinguishing waves from the motion of material.

SUCCESS CRITERIA - I can...

1. Identify the medium carrying the wave.
2. Use speed and distance to estimate travel time.
3. Explain why a simple travel time does not predict damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “medium” and “wave” to compare your model or result. Sentence starters are optional.

medium: the material through which a mechanical wave travels

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

01 Warm-up: bring back an earlier idea

A bundle of parallel rays passes through a lens and becomes 3 times as wide farther beyond it, with rays spreading immediately after the lens.
Identify the lens type and sketch the rays.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

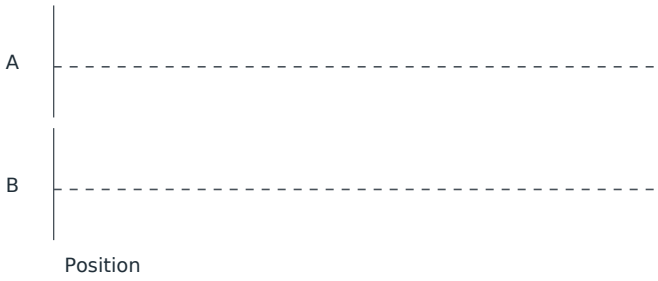
02 Worked example: explain the model

Reference: $d = v\,\Delta t$; $\Delta t = d/v$.
Use the supplied wave speed for its medium and wave type. These classroom models are not real hazard forecasts.

Worked reasoning: In a simple model, a wave amplitude falls from 4 to 2 units while intensity is proportional to amplitude squared. Intensity becomes one quarter, not one half.

Your turn: Draw the two waves and two corresponding relative-intensity bars.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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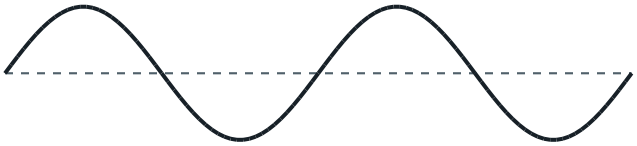
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw a disturbance traveling through water and mark local water motion. Why does wave arrival not mean the original water crossed the whole ocean?

Sentence starter: My claim is … . The diagram/data show …, so … .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

In a water-wave simulation, a disturbance travels at 20 m/s between sensors 200 m apart.
Calculate arrival time.
Mark the one-way path and explain why this is not the speed of an individual water particle.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why does knowing wave arrival time alone not establish local damage?
Name other evidence.

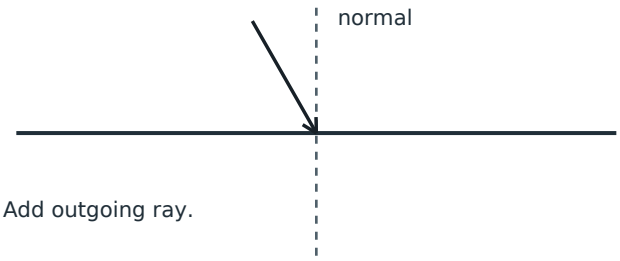
Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave enters a medium where its speed is 4 times as large, at a small angle below critical conditions. Does the transmitted ray bend toward or away from the normal?
Label the media, normal and ray directions.

Sentence starter: The … in my model represents … . Its label shows … .

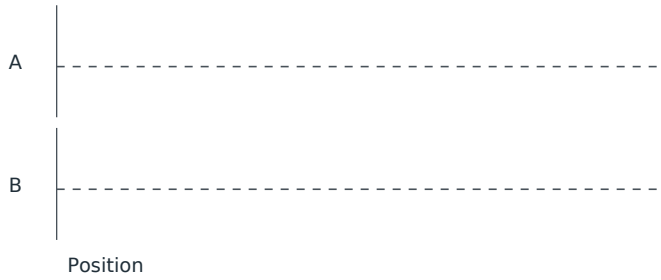


Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, distance doubles. How does travel time change?

Sentence starter: I notice … . This supports … because … .

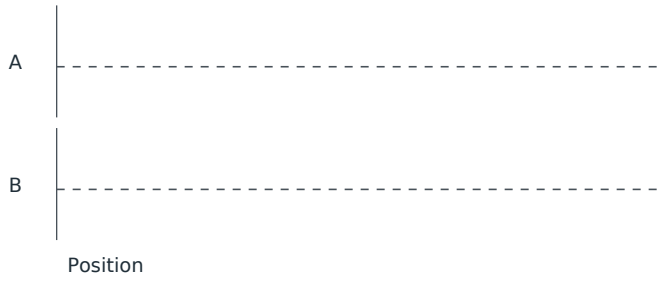


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two sites have amplitudes 3 and 1 units.
Compare relative intensities and name one real-world factor besides distance that could affect the readings.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3